

Introduction to Hosting Methods

In cloud computing, **hosting** refers to the method used to store and run applications, websites, and data so that users can access them over a network (usually the Internet).

There are mainly three common hosting methods:

1. **On-Premises Hosting**
2. **Cloud Hosting**
3. **Hybrid Hosting**

Each method differs in terms of **ownership, cost, maintenance, scalability, and control**.

Real-Life Example

Imagine a university managing its student management system.

- If the university installs servers inside its own campus IT room → **On-Premises Hosting**
- If the university uses Amazon Web Services (AWS) or Microsoft Azure → **Cloud Hosting**
- If it keeps sensitive data on campus but uses cloud for website hosting → **Hybrid Hosting**

This helps students understand hosting in practical terms.

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1 Cloud Hosting

Definition:

Cloud hosting is a method where computing resources (servers, storage, networking) are **distributed across the internet** and shared among multiple users.

Explanation:

- Resources are not located in one single physical place.
- They are stored in large data centers managed by providers like **AWS, Microsoft Azure, Google Cloud**.
- Uses virtualization technology.

- Follows a **pay-as-you-go** pricing model.

Example:

A small business hosting its website on Amazon Web Services (AWS).

Key Point:

- ✓ Scalable
 - ✓ Flexible
 - ✓ No need to buy hardware
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2 Dedicated Hosting

Definition:

Dedicated hosting means **renting a physical server exclusively** for your organization from a third-party provider.

Explanation:

- The server is not shared with other customers.
- You have full control over the server configuration.
- The hardware is owned by the provider but used only by you.

Example:

An e-commerce company renting a dedicated server from GoDaddy or HostGator.

Key Point:

- ✓ High performance
 - ✓ Better security than shared hosting
 - ✓ More expensive than cloud/shared hosting
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3 On-Premises Hosting

Definition:

On-premises hosting means hosting servers within your own organization's physical location, fully owned and managed by you.

Explanation:

- Servers are installed inside your office or building.
- You are responsible for maintenance, security, power backup, and cooling.
- Requires IT staff.

Example:

A bank storing all customer transaction data on servers inside its own building.

Key Point:

- ✓ Maximum control
- ✓ High initial cost
- ✗ Limited scalability

Comparison

Feature	Cloud Hosting	Dedicated Hosting	On-Premises Hosting
Ownership	Provider	Provider	Organization
Resource Sharing	Shared	Not Shared	Not Shared
Initial Cost	Low	Medium	High
Maintenance	Provider	Provider (mostly)	Organization
Scalability	High	Moderate	Limited